APT-BENCH: PATIENT-DISJOINT EVALUATION REVEALS GRANULARITY AND MEASUREMENT GAPS IN SINGLE-CELL APTAMER PROFILING

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ABSTRACT

Single-cell assays produce hundreds of thousands of cells, but the effective sample size for clinical prediction is determined by independent subjects. We introduce APT-Bench, a patient-disjoint benchmark of 361,792 single-cell aptamer profiles from 40 patients across six disease groups. It separates five-lineage and 27-subtype cell annotation from patient-level disease recognition and measurement-efficiency analysis. Across neural and classical models, coarse lineage recognition is consistently easier than fine subtype prediction. In this cohort, subtype F1 is more strongly associated with cell abundance than patient coverage, although coverage has limited variation and neither association establishes causality; most errors also cross lineage boundaries, exposing a persistent granularity gap. Cell-typing errors propagate downstream: disease macro-F1 from subtype composition falls from 0.572 to 0.400 when gold labels are replaced by out-of-fold predictions. Stronger direct APT discrimination is accompanied by patient-specific absolute-expression signatures that cannot be separated into biological and technical components with the available metadata. Under outer-test-isolated development selection, every fold chooses $B^* = 10$; the resulting 10-aptamer panels match the observed within-cohort performance of the full 293-aptamer panel. Random-panel and cross-model attribution controls indicate that this discriminative signal is distributed across alternative feature subsets. APT-Bench provides a statistically explicit setting for multiresolution, measurement-efficient single-cell learning, while limiting its disease conclusions to this cohort pending external validation.

1 INTRODUCTION

Cancer screening and diagnosis are usually framed at the patient level: does a person have disease, and if so which disease category is most likely? Single-cell liquid biopsy changes this framing. A single blood draw can support both a patient-level diagnostic decision and a cell-level characterization of the circulating immune compartment. These outputs are related, but they do not live at the same observational unit. Disease is a property of the patient, whereas lineage and subtype are properties of the cell.

This mismatch matters for evaluation. Our cohort contains more than 370,000 single cells, but only 40 patients. Treating cells as independent disease observations would substantially overstate the effective sample size of the diagnostic task. It would also hide a second asymmetry: lineage and subtype form a genuine cell-level hierarchy, but disease does not sit above that hierarchy. The clinically relevant problem is therefore not a single three-level taxonomy. It is a multi-resolution setting in which different targets require different evaluation protocols, inductive biases, and interpretations.

We study aptamer-based single-cell profiling. Apt-seq uses sequencable nucleic-acid affinity reagents to quantify molecules such as surface proteins while retaining cell barcodes and paired transcriptomic state (Wu et al., 2025). Unlike gene-vocabulary scRNA-seq or antibody-derived-tag integration benchmarks, our predictive input is a fixed 293-aptamer binding panel. This enables three questions: how performance changes from coarse lineage to fine subtype; how patient-level disease evaluation handles many correlated cells but only 40 subjects; and how many measured aptamers and held-out cells preserve that disease signal at inference.

APT-Bench fixes three patient-disjoint tracks: coarse-to-fine cell typing, patient-level disease recognition with statistical-unit audits, and measurement efficiency over aptamer-measurement and inference-aggregation budgets. Its reusable contribution is not a claim that one cohort spans deployment, but a common contract for a new modality: immutable patient folds, explicit observational units, leakage-resistant metrics, and outer-test-isolated budget controls that can extend to future clinical single-cell cohorts.

To instantiate the hierarchy track, we use DropCascade, a differentiable lineage-subtype cascade. The model combines a shared APT encoder, soft lineage routing, and subtype experts. We compare it with classical models and encoder-matched flat and HCE-trained Transformer controls. It serves as a hierarchy-aware neural reference rather than a claimed universal winner; the matched controls test whether any gain survives control of encoder capacity and training budget.

This perspective leads to three empirical findings. First, coarse lineage recognition is substantially easier than fine subtype recognition across all models. In an exploratory subtype audit, F1 is more strongly associated with cell abundance than patient coverage in this cohort, although coverage has limited variation and neither association establishes causality; most errors also cross the true lineage rather than remaining among biological siblings. Second, cell-typing error propagates to patient characterization: replacing gold subtype labels with OOF predictions reduces composition-based disease macro-F1 from 0.572 to 0.400. Direct disease prediction remains much stronger but depends on patient-specific absolute-expression structure, making overclaiming inappropriate without acquisition-batch metadata. Third, the disease signal is measurement-efficient but distributed: development-only selection preserves patient-level performance with 10 aptamers, including over 50 repeated inference subsamples at 100 held-out cells per patient, while 100-panel random controls and an XGBoost-SHAP attribution control argue against a unique indispensable biomarker set.

Our contributions are therefore as follows.

- **APT-Bench, a patient-disjoint reference benchmark for single-cell APT liquid biopsy.** We formalize three tasks with fixed patient folds, common inputs and labels, and a portable evaluation contract spanning hierarchy prediction, patient-level disease audit, and measurement efficiency.
- **A subtype-centered account of the granularity gap.** Beyond aggregate macro-F1, we quantify how cell abundance and patient coverage relate to each subtype's performance, decompose incorrect subtype predictions into sibling and cross-lineage errors, and report the coarse-to-fine gap for every model. We use DropCascade as a hierarchy-aware reference model rather than claiming universal superiority.
- **A bridge from cell typing to disease characterization.** Gold-versus-OOF-predicted composition isolates downstream error propagation; direct APT, confounding, development-only panel-selection, and inference-time equal-cell audits then distinguish within-cohort disease signal from clinical evidence.

2 Related Work

Liquid biopsy and multi-cancer detection. Blood-based multi-cancer early detection has focused primarily on circulating tumor DNA, methylation, fragmentomics, and bulk protein assays (Cohen et al., 2018; Zeng et al., 2026; Wan et al., 2025; Kahwati et al., 2025). These approaches target a patient-level diagnostic decision. Our setting differs because APT profiling observes individual circulating cells, which introduces a second prediction problem at cell resolution: identifying lineage and fine subtype structure within the same blood draw. The resulting task is not simply "more labels," but a change in statistical unit and hierarchy structure.

Single-cell foundation models and modality mismatch. Large pretrained models such as Geneformer, scGPT, and scFoundation learn representations from transcriptomic data and are widely used as reference points for single-cell representation learning (Theodoris et al., 2023; Cui et al., 2024; Hao et al., 2024). Their success does not transfer automatically to our setting. The input modality is different, since APT consists of a fixed 293-dimensional protein-oriented panel rather than a gene-vocabulary sequence. More importantly, recent evaluation work shows that frozen or zero-shot single-cell foundation-model embeddings are often matched by simpler baselines once the downstream task and evaluation protocol are made explicit (Kedzierska et al., 2025). This motivates training directly on APT profiles and treating evaluation design itself as a first-class contribution.

Reusable single-cell benchmarks. Open Problems separates datasets, methods, and metrics behind public loaders and evaluation code (Luecken et al., 2025); a 2025 multimodal benchmark similarly releases processed RNA/ADT/ATAC inputs and task-specific pipelines (Liu et al., 2025). SPARE standardizes sample-representation interfaces (Shitov et al., 2025), and PaSCient evaluates patient representations with study-separated partitions (Liu et al., 2026). None combines a fixed aptamer-binding panel, coarse-to-fine cell typing, patient-level disease audit, and aptamer-measurement/inference-aggregation budgets. APT-Bench targets this narrower gap rather than replacing broad RNA or integration benchmarks.

Hierarchical cell-type classification. Cell identity naturally admits coarse-to-fine structure, and recent work has explored losses or architectures that exploit this structure for annotation and out-of-distribution robustness (Jia et al., 2025; Cultrera di Montesano et al., 2026). Our use of hierarchy is narrower and more explicit: disease is kept outside the hierarchy, while lineage and subtype form the only genuine taxonomy. We compare DropCascade not only with flat classifiers but also with a matched flat FT-style Transformer control (Gorishniy et al., 2021) and the same control trained with a two-level adaptation of hierarchical cross-entropy (HCE), the most directly related hierarchy-aware loss.

Measurement-efficient disease recognition. Clinical translation of high-plex assays depends not only on accuracy but also on measurement cost. APT-Bench therefore asks how many aptamers are needed under outer-test-isolated panel selection and how many held-out-cell predictions suffice for patient aggregation. The latter is an inference sensitivity, not reduced-cell training. This is distinct from biomarker discovery: our goal is to quantify predictive budget and redundancy in the assay, not to claim that a small panel is a universal causal signature.

Tabular deep learning versus classical machine learning. APT expression is a medium-dimensional tabular signal. In that regime, gradient-boosted trees remain strong defaults and often rival or outperform deep tabular models (Grinsztajn et al., 2022; Erickson et al., 2026). This literature is directly reflected in our results: XGBoost and encoder-matched Transformer controls remain competitive at both resolutions, so DropCascade cannot be justified by a blanket accuracy claim. We instead use it as a structured reference for the hierarchy diagnostics, while disease recognition is audited with simple, strong tabular baselines.

3 Problem Setup

3.1 Dataset and Prediction Targets

Cohort construction and specimen source. APT-Bench is derived from peripheral-blood clinical specimens collected from 40 participants: patients with colorectal cancer (CRC), gastric cancer (GC), hepatocellular carcinoma (HCC), pancreatic cancer (PC), or biliary tract cancer (BTC), and healthy controls (Normal). The six groups contain 6–8 participants each. Clinical specimens were processed into single-cell suspensions before assay preparation. The study workflow documents quality control using the forward/side-scatter cell profile and viable-cell proportion before library construction.

Paired single-cell assay and label construction. The assay jointly measures transcriptomic state and binding of 293 target-specific nucleic-acid aptamers on the same cells. Transcriptome and aptamer-barcode libraries are constructed separately and linked at the cell level by their shared barcodes. Cell clustering and biological annotation are performed from the transcriptomic modality; these annotations define the lineage and subtype targets used by APT-Bench. Disease labels come from the participant-level cohort assignment. The benchmark input contains only the 293 APT measurements, so transcriptomic measurements are used to construct labels but are never exposed to a prediction model.

TODO: Complete the cohort, specimen-processing, assay, quality-control, and annotation fields listed in Appendix A.1; report unavailable metadata explicitly rather than leaving them implicit.

Benchmark cohort. The raw cohort contains 374,854 cells. After excluding the transcriptome-derived *Unknown* subtype label, which denotes uncertain annotation rather than a validated biological state, the main benchmark contains 361,792 cells. Each cell is represented by a 293-dimensional APT expression vector. Expression values are log-transformed and standardized feature-wise using statistics estimated from the training partition only. The 293 aptamers are designed probes rather than an arbitrary learned basis.

APT-Bench contains two kinds of targets at different observational units.

- **Patient-level disease recognition.** Each patient is assigned to one of six disease groups.
- **Cell-level hierarchy prediction.** Each cell is assigned to one of five coarse immune lineages and one of 27 fine subtypes. Every subtype maps to exactly one parent lineage through a manually curated lookup table.

Disease is therefore not treated as the root of a three-level hierarchy. It is a separate patient-level label observed on the same blood draw.

3.2 Benchmark Contract and Extensibility

APT-Bench fixes the unit of input, prediction target, split, and metric for each track: five-lineage and 27-subtype prediction from one APT vector per cell; six-class disease prediction from all cells of one participant; and outer-test-isolated aptamer-measurement plus inference-time cell-budget evaluation of that participant-level task. Novelty is not part of the primary contract. The machine-readable manifest at `benchmark/splits/patient_folds.json` assigns every pseudonymous patient to one outer fold; all selection remains inside development patients. Portable components include patient-clustered uncertainty, hierarchy diagnostics, pseudoreplication and fingerprint audits, and outer-test-isolated budget curves. The full interface and release contract appear in Appendix A.2. This cohort cannot estimate cross-site or cross-platform generalization.

TODO: Deposit the versioned data and executable evaluation release specified in Appendix A.2; otherwise describe the work as a dataset case study rather than a reusable benchmark.

3.3 Evaluation Protocols

We evaluate models under two main protocols with different purposes.

Patient-level 5-fold cross-validation. This is the primary protocol for all headline claims. Patients are partitioned into five disease-stratified folds, each containing eight held-out test patients. At every fold, preprocessing statistics, model fitting, and hyperparameter selection use only the training and validation patients from the remaining folds. We pool out-of-fold predictions across all five test folds, yielding a patient-disjoint evaluation spanning all 40 patients.

Cell-level split control. As a protocol-sensitivity control, cells are randomly split irrespective of patient identity. Cells from the same patient can therefore appear in both training and evaluation sets. We do not treat this as a realistic deployment protocol. Its only role is to quantify performance inflation caused by leakage.

3.4 Class Distribution

Because the subtype distribution is long-tailed, we use macro-F1 as the primary metric for coarse-lineage and fine-subtype prediction. Accuracy is secondary. Primary point estimates pool cells and are therefore cell-weighted despite patient-disjoint splits; Appendix A.5 reports equal-cell-per-patient sensitivity results. For uncertainty, we compute 95% confidence intervals from 2,000 paired patient-level bootstrap resamples over the 40 pooled test patients. The primary inferential family contains DropCascade versus HCE and versus XGBoost at the coarse and fine endpoints. Two-sided bootstrap p-values use twice the smaller tail probability with a plus-one correction and are Holm-adjusted across these four tests; comparisons outside this family are exploratory.

From the same pooled predictions, joint hierarchy metrics are exact-path accuracy, example-averaged lineage/subtype F1, and subtype-tree distance; support correlations, sibling errors, and the coarse–fine gap are exploratory difficulty diagnostics.

We report disease recognition at the patient level, since the clinical decision is made per patient rather than per cell. Cell-level disease accuracies are included only as supplementary evidence and are not interpreted as though hundreds of thousands of cells were independent disease observations.

4 Hierarchy-Aware Reference Model

APT-Bench is intentionally broader than any single model. We nevertheless need a hierarchy-aware reference architecture that can operate directly on APT profiles under the benchmark protocols. DropCascade plays this role. It focuses on the genuine cell-level hierarchy, lineage $\rightarrow$ subtype, and leaves disease recognition to separate tabular baselines and patient-level aggregation. We do not present the architecture as a primary contribution: without full five-fold component ablations, its global path, coarse auxiliary objective, soft routing, and contrastive construction should be read as implementation choices rather than independently validated sources of performance.

4.1 Cell Encoder

Each cell is represented by a standardized APT vector $x \in \mathbb{R}^{293}$. We embed feature identity and feature value separately. For aptamer index i, we form a token

$$t_i = \mathrm{Emb}(i) + \mathrm{MLP}_{\mathrm{val}}(x_i), \qquad i = 1, \ldots, 293. \tag{1}$$

A learnable `[CLS]` token is prepended to the resulting sequence and processed by a pre-norm Transformer encoder with 4 layers, 8 attention heads, hidden dimension 256, feed-forward dimension 512, GELU activations, and dropout 0.1. The final `[CLS]` representation

$$z = \mathrm{Encode}(x) \in \mathbb{R}^{256} \tag{2}$$

serves as the shared cell representation for downstream tasks.

4.2 Soft-Routed Hierarchical Cascade

Let $M = 5$ be the number of coarse lineages and $K = 27$ the number of fine subtypes. Each subtype k belongs to exactly one parent lineage $m(k)$. A coarse head produces lineage logits

$$c = W_c z \in \mathbb{R}^M, \tag{3}$$

with routing distribution

$$p(m \mid x) = \mathrm{softmax}(c/\tau)_m, \tag{4}$$

where $\tau = 1$.

Instead of a hard cascade, which would force subtype prediction to follow only the top predicted lineage, DropCascade uses a soft mixture of a global subtype path and lineage-specific experts:

$$s_k(x) = \underbrace{g_k(z)}_{\text{global path}} + \alpha \, p(m(k) \mid x) \, e_{m(k),k}(z) + \gamma \log\big(\epsilon + (1 - \epsilon) p(m(k) \mid x)\big), \tag{5}$$

Here $g(z)$ is a global subtype head, $e_{m,k}(z)$ is the expert associated with lineage m, $\alpha = 1.0$, $\epsilon = 0.1$, and γ is ramped from 0 to 0.5 during early training. The global path retains a direct subtype-logit contribution alongside the routed expert contribution; we do not isolate its effect from the other components. Final predictions are

$$\hat{k} = \arg\max_k s_k(x), \qquad \hat{m} = \arg\max_m c_m. \tag{6}$$

4.3 CROSS-DISEASE CONTRASTIVE REGULARIZATION

The reference model includes a cross-disease supervised contrastive term. Let

$$u = \frac{\mathrm{Proj}(z)}{\|\mathrm{Proj}(z)\|_2} \tag{7}$$

be a normalized projection of the encoder representation. For anchor cell i, positives are cells with the same subtype but a different disease label. Same-subtype cells from the same disease are excluded by construction. Cells from different subtypes are treated as negatives, with additional emphasis on negatives that share the same coarse lineage. The contrastive loss weight is ramped from 0 to 0.2 after the early warmup period. Because we do not compare this construction with standard supervised contrastive learning across all five folds, we make no claim that the cross-disease positive rule independently improves generalization.

4.4 OPTIMIZATION AND MODEL SCOPE

The classification cascade is trained jointly using

$$\mathcal{L} = \lambda_{\text{fine}}\mathcal{L}_{\text{fine}} + \lambda_{\text{coarse}}\mathcal{L}_{\text{coarse}} + w_{\text{contrast}}(t)\mathcal{L}_{\text{contrast}}, \tag{8}$$

where $\mathcal{L}_{\text{fine}}$ and $\mathcal{L}_{\text{coarse}}$ are cross-entropy losses with $\lambda_{\text{fine}} = 1.0$ and $\lambda_{\text{coarse}} = 0.5$. The contrastive weight increases to 0.2 according to the schedule in §4.3.

We optimize the cascade using AdamW with learning rate 10^{-4} and weight decay 10^{-5}, gradient clipping at norm 1.0, batch size 512, and mixed-precision training. Models are trained for at most 50 epochs with early stopping based on validation fine-level macro-F1 with patience 10. The final configuration does not use class-balanced sampling, as our experiments found that it did not improve performance after correcting an implementation issue; additional details are provided in Appendix A.10.

Matched Transformer controls. To isolate encoder capacity from hierarchical modeling, a flat FT-style Transformer control (Gorishniy et al., 2021) removes the cascade while retaining the identical APT tokenizer, Transformer dimensions, optimizer, early stopping, and training budget above. We call it FT-style because its tokenizer is exactly the APT feature tokenizer used by DropCASCADE, rather than the original numerical/categorical FT-Transformer tokenizer. This model and its HCE counterpart predict the same $M + K$ lineage-and-subtype nodes. The flat model applies independent cross-entropy to the lineage and subtype logit slices. For HCE (Cultrera di Montesano et al., 2026), let $R_{ij} = 1$ when node j is node i or its descendant, $p = \mathrm{softmax}(q)$ over all nodes, and $s = Rp$. We minimize $-\lambda_{\text{fine}} \log s_{M+k} - \lambda_{\text{coarse}} \log s_m$ for observed subtype k and lineage m. Because every fine label here is a leaf, the fine HCE term alone would reduce to ordinary CE; including the simultaneously observed parent label makes this a stated two-level adaptation rather than an unqualified reproduction of the mixed-granularity setting in the original work. All three Transformer models are evaluated from the checkpoint selected by patient-validation fine macro-F1, without an additional post-selection calibration stage.

Disease recognition is not handled by a dedicated head inside DropCASCADE. This is a deliberate modeling decision rather than a claim that disease is solved in a strong external sense: within-cohort disease prediction is already nearly saturated by simple classical baselines, while the architectural challenge addressed by DropCASCADE is the unresolved cell-level hierarchy problem.

5 RESULTS

5.1 DISEASE RECOGNITION IS STRONG WITHIN COHORT, BUT REQUIRES CAVEATS

Under the fixed patient-level folds, five classical cell-level baselines all yield the same majority-vote result: $40/40$ patients are correctly assigned (95% CI $[0.912, 1.000]$; Table 1). A genuinely patient-level median-APT model reaches accuracy 0.950 and macro-AUROC 1.000; none of 1,000 patient-label permutations matches its accuracy (plus-one $p = 0.000999$; Appendix A.3).

Aggregation / Representation	**Patient Acc.**	**Macro-F1**	**Macro-AUROC**
Cell-level classifiers*, majority vote	**1.000**	**1.000**	–
Patient-level APT median + LR†	0.950	0.948	1.000
Gold subtype composition + LR‡	0.575	0.572	0.842
OOF subtype composition + LR§	0.400	0.400	0.749
Gold lineage composition + LR‡	0.400	0.377	0.678
OOF lineage composition + LR§	0.300	0.282	0.627

* Five cell-level classifiers share the $40/40$ majority-vote result; their distinct cell metrics appear in Table 4. † One 293-APT median vector per patient. ‡ Paired-transcriptomic gold labels; § patient-disjoint DropCascade OOF labels. Composition rows use CLR (pseudocount 10^{-4}), the fixed folds, and prespecified LR $C = 0.1$.

Table 1: **Patient-level** disease classification ($n = 40$). The majority-vote accuracy has 95% Clopper–Pearson CI $[0.912, 1.000]$. Replacing gold cell labels with patient-disjoint OOF predictions degrades every composition endpoint, linking cell-typing error to downstream characterization. All disease results are within-cohort estimates, not evidence of clinical generalization; see Appendix A.3.

Composition directly connects cell typing to patient characterization. Replacing gold labels with strictly OOF DropCascade predictions reduces subtype-composition macro-F1/AUROC from $0.572/0.842$ to $0.400/0.749$ and lineage-composition macro-F1 from 0.377 to 0.282. Typing errors therefore propagate downstream, while all composition variants remain well below median APT, indicating that absolute APT structure carries the stronger within-cohort disease signal.

These results do not justify a claim of solved clinical diagnosis. Sensitivity analyses in Appendix A.3 show that the disease signal depends strongly on patient-specific absolute-expression structure: patient-wise centering reduces patient-level accuracy from 1.000 to 0.650, and a 40-way patient-identity diagnostic reaches 88.4% accuracy on held-out cells. Because acquisition-batch metadata are unavailable, the observed offsets cannot be cleanly decomposed into biological versus technical sources. We therefore use disease recognition primarily to motivate the need for careful statistical-unit accounting in APT-Bench, not as the main architectural success story.

5.2 Patient-Disjoint Cell-Level Hierarchy Prediction Remains Difficult

The harder unresolved problem is cell-level hierarchy prediction under patient-disjoint evaluation. Table 2 shows that coarse lineage recognition is consistently easier than fine subtype recognition. For DropCascade, coarse macro-F1 reaches 0.326 while fine macro-F1 is only 0.152; for the matched flat FT-style Transformer, the corresponding values are 0.323 and 0.147. This granularity gap appears across model families and indicates that APT captures broad immune identity more reliably than fine subtype structure.

The matched comparison changes the architectural conclusion. Relative to the flat Transformer, DropCascade differs by only $+0.0036$ coarse macro-F1 (paired patient-bootstrap 95% CI $[-0.0078, 0.0146]$) and $+0.0055$ fine macro-F1 (95% CI $[-0.0021, 0.0125]$). HCE has the highest coarse point estimate (0.333), but its difference from DropCascade is also unsupported: DropCascade$-$HCE is -0.0069 (95% CI $[-0.0154, 0.0026]$) at the coarse level and $+0.0015$ (95% CI $[-0.0083, 0.0047]$) at the fine level. Differences from XGBoost likewise do not exclude zero. Thus the experiment does not isolate a performance gain from the cascade: the defensible result is the persistent coarse-to-fine difficulty under patient-disjoint evaluation, not superiority of a routing mechanism.

Exact-path accuracy is only 0.248–0.259; HCE has the highest value, whereas XGBoost has the highest hierarchical F1 (0.420) and lowest tree distance (2.375). Natural consistency remains in Appendix A.8 because constrained decoding can alter it without improving predictive quality.

5.3 Why 361,792 Cells Still Yield Only 0.15 Subtype Macro-F1

Subtype support associations are exploratory. For subtype k, marginal $\mathrm{Spearman}(F_{1,k}, \log n_k)$ versus $\mathrm{Spearman}(F_{1,k}, P_k)$ is 0.486/0.288 for DropCascade and 0.540/0.275 for XGBoost. After

Model	Coarse M-F1	Fine M-F1	Fine Acc.	Exact Path	Hier. F1	Tree Dist.↓
DropCascade	0.326	**0.152**	0.305	0.253	0.404	2.431
Flat FT-style Transformer	0.323	0.147	0.315	0.257	0.411	2.407
FT-style Transformer + HCE	**0.333**$^{\text{n.s.}}$	0.151$^{\text{n.s.}}$	0.306	**0.259**	0.405	2.418
XGBoost	0.322$^{\text{n.s.}}$	0.143$^{\text{n.s.}}$	**0.323**	0.248	**0.420**	**2.375**

Table 2: Patient-disjoint 5-fold hierarchy evaluation on the same pooled out-of-fold predictions from 40 patients. Exact Path requires both lineage and subtype to be correct. Hierarchical F1 is the example-averaged F1 between the two-node true and predicted sets $\{\text{lineage}, \text{subtype}\}$. Tree Dist. is the mean subtype-tree edge distance: 0 for the correct subtype, 2 for a sibling, and 4 for a cross-lineage prediction (lower is better). The primary comparisons are DropCascade versus HCE and versus XGBoost for coarse and fine macro-F1; their two-sided tests use 2,000 paired patient bootstrap resamples with Holm correction across four tests. Other comparisons and the three hierarchy metrics are descriptive. $^{\text{n.s.}}$ denotes no Holm-adjusted significance; coarse–fine consistency remains in Appendix A.8.

rank residualization, $\rho_{c|p}$ is 0.411/0.496 and $\rho_{p|c}$ is $-0.051/-0.130$. With only 27 subtypes and 16 at the 40-patient coverage ceiling, these are exploratory associations: they establish neither causality nor a significant difference. Appendix A.4 provides plots and patient-clustered F1 intervals for every subtype.

Raw error structure is dominated by cross-lineage predictions. Only 25.0% of DropCascade errors and 24.6% of XGBoost errors retain the true parent; across models, 70.5–80.9% cross lineage (Table 3). Because this rate depends on class marginals, it is descriptive rather than evidence of hierarchy awareness; Appendix A.4 provides fixed-marginal null comparisons.

The granularity gap is a benchmark finding. For each model, $G = F_1^{\text{coarse}} - F_1^{\text{fine}}$ is positive and ranges from 0.099 to 0.200. We use G only to expose persistent coarse-to-fine difficulty, not as a new metric contribution.

Model	ρ_c	ρ_p	$\rho_{c\|p}$	$\rho_{p\|c}$	G	R_{sibling}	Cross-lineage
DropCascade	0.486	0.288	0.411	-0.051	0.174	0.250	0.750
Flat FT-style Transformer	0.505	0.255	0.464	-0.123	0.176	0.243	0.757
FT-style Transformer + HCE	0.513	0.318	0.427	-0.034	0.183	0.257	0.743
XGBoost	0.540	0.275	0.496	-0.130	0.179	0.246	0.754
Logistic Regression	0.435	0.107	0.488	-0.267	0.167	0.295	0.705
Linear SVM	0.339	-0.012	0.463	-0.335	0.200	0.265	0.735
Random Forest	0.661	0.418	0.564	-0.035	0.158	0.191	0.809
Reference Correlation	0.376	0.003	0.499	-0.354	0.099	0.225	0.775

Table 3: Exploratory subtype diagnostics on pooled patient-disjoint predictions. ρ_c and ρ_p are marginal Spearman associations of subtype F1 with log cell count and patient coverage; $\rho_{c|p}$ and $\rho_{p|c}$ are partial Spearman correlations computed by residualizing rank-transformed variables. $G = F_1^{\text{coarse}} - F_1^{\text{fine}}$. Among errors, R_{sibling} retains the true parent lineage and Cross-lineage is its complement. With only 27 subtypes and limited variation in patient coverage, these associations are descriptive and do not establish causality.

5.4 Protocol Sensitivity: Patient-Level versus Cell-Level Splitting

Cell-level splitting inflates coarse/fine macro-F1 by $+0.090/+0.108$ for DropCascade and $+0.050/+0.069$ for XGBoost (Appendix A.5). APT-Bench therefore fixes patient-disjoint evaluation rather than treating the split as a reporting detail.

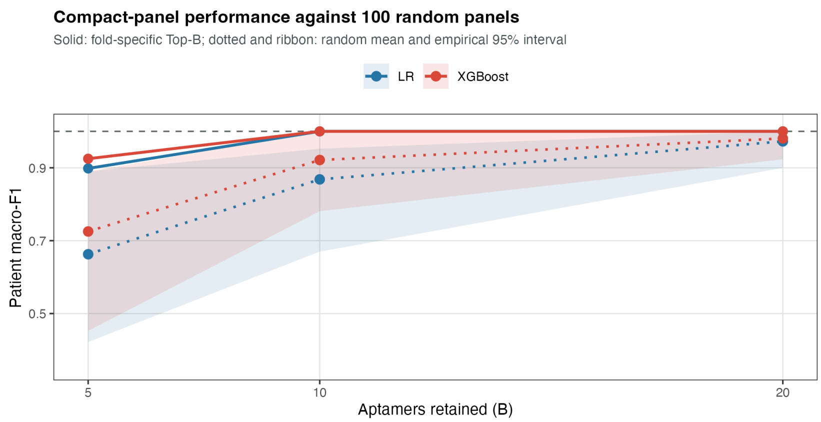

Figure 1: Patient-level disease macro-F1 under compact aptamer-budget controls. Solid lines show fold-specific Top-B panels; dotted lines and ribbons show the mean and empirical 95% interval over 100 random-panel repeats. Each repeat shares one panel across outer folds, with models refit foldwise. The dashed line is the full 293-aptamer reference; scores pool the 40 outer-test patients once each.

5.5 Disease Signal Is Measurement-Efficient but Predictively Distributed

The compact-panel analysis in Appendix A.9 provides a final perspective on what the APT modality supports. Under outer-test-isolated, development-only budget selection, all five folds choose $B^* = 10$, and the resulting 10-aptamer panels reproduce the full-panel patient macro-F1 of 1.0 for both Logistic Regression and XGBoost on every outer test fold. An LR inference-time sensitivity obtains the same macro-F1 across 50 subsamples at 100 held-out cells per patient; model training still uses all outer-development cells. Across 100 Random-B panel repeats, performance also rises quickly with B; moreover, LR-coefficient and XGBoost-SHAP Top-10 panels overlap only moderately but both retain perfect patient-level performance. The result therefore supports reduced aptamer measurement and inference aggregation budgets, not training with 100 cells or a unique minimal biomarker set.

Compact-panel recovery does not resolve whether the disease signal is biological, technical, or mixed. It instead shows a central asymmetry: within-cohort disease recognition is statistically and operationally easier than patient-disjoint cell-level hierarchy prediction.

6 Discussion

APT-Bench reveals a coarse–fine gap whose errors propagate to composition-based patient characterization; matched controls show no significant DropCascade advantage. Disease scores are centering-sensitive, and compact panels establish within-cohort measurement efficiency rather than causal biomarkers. This single-site, 40-patient cohort lacks batch metadata and external validation, so novelty remains exploratory. Its reusable contribution is the fixed schema, splits, tasks, and evaluator (Luecken et al., 2025; Liu et al., 2025); clinical claims require cross-site cohorts.

7 Conclusion

APT-Bench defines linked patient-disjoint hierarchy typing, disease audit, and measurement-efficiency tasks for single-cell APT profiles. External cohorts remain necessary for clinical claims.

AI use statement

TODO: **Required for ICLR 2027.** Disclose every generative-AI tool and model used for writing, editing, coding, analysis, literature review, or figure preparation; identify the tasks for which each tool was used, describe human verification of AI-assisted outputs, and state that the authors take responsibility for the final text, claims, code, and artifacts.

Ethics statement

TODO: State the human-subjects approval or exemption and informed-consent status; describe data de-identification, access restrictions, privacy protections, intended clinical/research use, foreseeable misuse or bias risks, and any conflicts of interest. Do not include identifying institutional details if they would break anonymity.

Reproducibility statement

TODO: Provide the anonymous code and data-access locations, exact patient-split manifest, preprocessing and evaluation commands, software environment, random seeds, model hyperparameters, compute requirements, and paths for the prediction artifacts and tables used in every reported result.

Author Contributions

TODO: Add author contributions using CRediT roles (for example: conceptualization, methodology, software, validation, formal analysis, data curation, visualization, writing, supervision, and funding acquisition). Preserve anonymity in the review version if required.

Acknowledgments

TODO: Add funding sources and grant numbers, computational resources, data or clinical contributors, and non-author assistance; verify whether any acknowledgment must be omitted or anonymized during double-blind review.

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A APPENDIX

A.1 DATASET CONSTRUCTION AND DOCUMENTATION CHECKLIST

The available study workflow establishes the high-level sequence used to construct APT-Bench: clinical peripheral-blood specimens from five cancer groups and healthy controls are converted to single-cell suspensions; cell-profile and viability quality control precedes a paired single-cell transcriptome and 293-aptamer assay; separate transcriptome and aptamer-barcode libraries are linked at the cell level; and transcriptome-derived clusters and annotations provide the cell labels. For a benchmark release, this overview must be accompanied by the following provenance and protocol fields.

- TODO: Report recruiting site(s) and dates, prospective versus retrospective collection, exact patient count per group, inclusion/exclusion and diagnostic criteria, and whether each participant contributed exactly one specimen.
- TODO: Report available demographics and clinical covariates, including age, sex, disease stage, treatment status, and comorbidities; summarize missingness and state which fields cannot be released.
- TODO: Report blood volume, collection tubes, time to processing, cell-isolation and cryopreservation procedures, quality-control thresholds, and sample/cell exclusion counts.
- TODO: Report assay/platform and reagent versions, sequencing instruments and depth, 293-probe panel design, and a verified protein-target manifest or explicitly state why only anonymized probe identifiers can be released.
- TODO: Report barcode matching, doublet removal, transcriptomic preprocessing, clustering, marker-based annotation and adjudication, APT normalization and filtering, all software versions and parameters, and whether annotators were blinded to patient and disease identity.
- TODO: Add a cohort-flow diagram with participant and cell counts after every exclusion step, and provide the ethics approval/consent, de-identification, data-access, license, and intended-use terms referenced in the Ethics and Reproducibility Statements.

A.2 BENCHMARK INTERFACE AND RELEASE CONTRACT

The hierarchy submission interface contains pseudonymous cell and patient IDs, true and predicted lineage, and true and predicted subtype. The disease interface contains one row per patient with the true disease and six class scores. The efficiency track additionally records the aptamer measurement budget, inference-time cell budget, selection seed, and the development-only feature ranking. These common interfaces permit new models to be scored without reproducing their internal training pipelines.

The evaluation layer is portable to other clinical single-cell datasets. It implements group-disjoint out-of-fold prediction; patient-clustered bootstrap uncertainty; coarse and fine macro-F1, exact-path accuracy, hierarchical F1, subtype-tree distance, granularity gap, and sibling versus cross-lineage error; patient-level aggregation versus cell-level pseudoreplication; patient-centering and identity diagnostics; and nested feature- and cell-budget curves. A future dataset can replace the assay features, hierarchy, or clinical endpoint while retaining these interfaces. This modular task–dataset–metric separation follows recent living single-cell benchmarks (Luecken et al., 2025); standardized metadata and programmatic access similarly enable cross-study reuse in CZ CELLxGENE (CZI Cell Science Program et al., 2025).

A complete release must contain the processed APT matrix; pseudonymous cell and patient identifiers; disease and cell labels; hierarchy and feature metadata; the immutable fold manifest; metric implementations; baseline configurations; and a single command that regenerates the headline tables from model predictions. TODO: Before submission, add data and code URLs, explicit licenses, file checksums, an environment lockfile, an archival DOI, and the end-to-end command. If participant-level data require controlled access, release the complete schema, all labels permitted by consent, the split manifest, evaluator, and access procedure.

A.3 DISEASE CLASSIFICATION: CELL-LEVEL SOURCE PREDICTION AND PATIENT-LEVEL SENSITIVITY ANALYSES

Cell-level prediction of the disease source of a held-out patient. Table 1 reports patient-level disease classification, the correct statistical unit for a per-patient diagnostic call. For reference and comparability with earlier reporting conventions, Table 4 additionally reports the pooled *cell-level* metric, in which every held-out cell is treated as an independent prediction target and pooled across all 361,792 cells from the five test folds. This is an easier and statistically different task than patient-level diagnosis, since cells from the same patient are correlated observations rather than independent draws and patients with more cells contribute more weight. We therefore report this metric only as a lower-level diagnostic.

Model	Accuracy	Macro-F1	Weighted-F1	Balanced Acc.
Logistic Regression	**0.991**	**0.989**	**0.991**	**0.989**
XGBoost	0.990	0.988	0.990	0.987
Linear SVM	0.988	0.986	0.988	0.985
Random Forest	0.963	0.955	0.963	0.953
Reference Correlation†	0.921	0.909	0.921	0.915

† SingleR-style reference-correlation classifier (Spearman nearest-centroid mapping), following the template-matching principle used by reference-mapping methods such as SingleR (Aran et al., 2019).

Table 4: **Cell-level** prediction of the disease source of a held-out patient's cells (6-class), pooled over all 361,792 out-of-fold cells from the patient-level 5-fold protocol. This is a different, easier statistical task than the patient-level diagnosis reported in Table 1: cells from the same patient are correlated observations rather than independent draws, and this metric implicitly weights patients with more cells more heavily. We report it here only as a cell-level reference point, not as a patient-level diagnostic result.

Patient-wise centering sensitivity. To test whether disease prediction relies on each patient's absolute expression level rather than only on within-patient relative structure, we repeated Logistic Regression training and evaluation after subtracting each patient's own per-aptamer median from every cell in both training and test data. This transformation preserves within-patient cell-to-cell variation while removing a patient-specific location shift. Cell-level accuracy drops from 0.991 to 0.401 (macro-F1 $0.989 \rightarrow 0.348$), and patient-level majority-vote accuracy drops from $40/40$ to $26/40$. This collapse shows that disease recognition depends substantially on patient-level absolute-expression structure, but it does not by itself determine whether that structure is biological, technical, or a mixture of both.

Patient-fingerprint diagnostic. We additionally trained a 40-way classifier to predict patient identity from individual cells, using a within-patient 50/50 cell split for this diagnostic only. The classifier identifies the correct patient from a single held-out cell with 88.4% accuracy (macro-F1 0.862), against a 2.5% chance level. Individual cells therefore carry a strong and easily recoverable patient-specific signature, consistent with the centering sensitivity above.

Patient-level summary-feature control. Finally, to confirm that the disease signal does not rely only on cell-level pseudo-replication, we trained a genuinely patient-level classifier using each patient's median 293-dimensional APT profile. This summary-feature model reaches patient accuracy 0.950, patient macro-F1 0.948, and macro-AUROC 1.000. Together with the sensitivity analyses above, this indicates that disease information is present at the patient level but strongly entangled with absolute-expression offsets that cannot be cleanly separated without acquisition-batch metadata.

Cell-typing error propagation to disease characterization. We connect the hierarchy and disease tracks by replacing the gold cell labels in the composition baseline with DropCASCADE predictions. These are the pooled patient-disjoint OOF predictions used in Table 2: every predicted label for patient p comes from a cell-typing model whose training and validation sets exclude p. For each patient, we aggregate either the gold or predicted labels into a 27-subtype or five-lineage proportion vector, apply the same CLR transform (pseudocount 10^{-4}), and fit the same patient-level multinomial LR ($C = 0.1$) under the fixed five folds. Predicted subtype composition reduces patient accuracy/macro-F1/AUROC from $0.575/0.572/0.842$ to $0.400/0.400/0.749$; predicted lineage composition reduces them from $0.400/0.377/0.678$ to $0.300/0.282/0.627$. The source cell labels and the downstream disease calls are therefore both out of sample for each test patient. These results quantify error propagation rather than train-set reconstruction: imperfect typing degrades patient characterization, while the stronger median-APT result indicates that composition is not the principal carrier of the observed disease separation.

Patient-label permutation control. We evaluated the same patient-level median-profile classifier under 1,000 label permutations. Disease labels were permuted across the 40 patients as indivisible units, preserving the disease-group counts $(6, 8, 7, 7, 6, 6)$; labels were never shuffled across cells. We retained the original five patient folds rather than restratifying after permutation. No permuted dataset reached the observed accuracy of 0.950 or macro-F1 of 0.948, giving plus-one permutation $p = (0 + 1)/(1000 + 1) = 0.000999$ for both statistics. The null accuracy mean was 0.137, with empirical 95% interval $[0.025, 0.275]$. This rejects a random patient-label explanation under the fixed protocol, but does not distinguish biological signal from patient-associated technical structure.

A.4 Per-Subtype Support and Performance Audit

Figure 2 visualizes the two marginal support–F1 associations for DropCascade and XGBoost. The vertical stack at 40 patients makes the coverage ceiling explicit. These plots and the partial correlations are exploratory: they neither establish causality nor test whether two dependent correlations differ significantly.

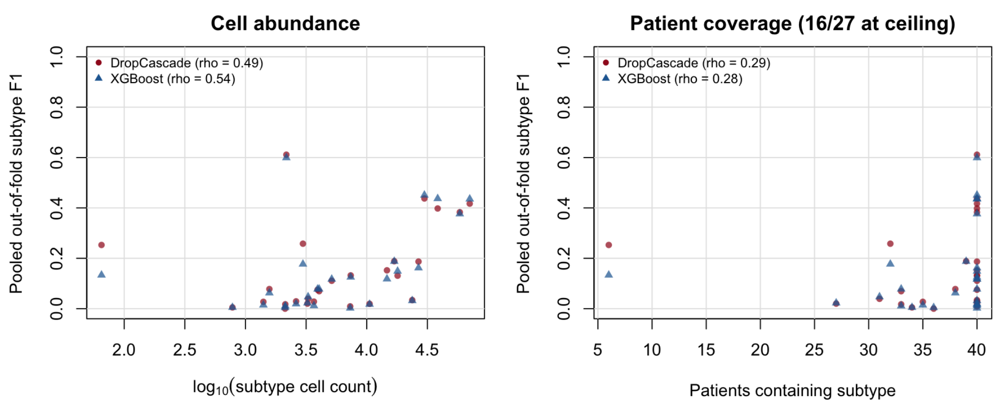

Figure 2: Per-subtype pooled out-of-fold F1 versus cell abundance (left) and patient coverage (right). Each marker is one of 27 subtypes; both models use the same ground-truth support values. Spearman correlations are marginal, and 16 of 27 subtypes occur in all 40 patients.

Table 5 gives all subtype-level quantities and 95% percentile intervals from 2,000 patient-clustered bootstrap resamples. Counts and coverage use pooled ground truth, while F1 uses the same patient-disjoint OOF predictions as Table 2. Mean cells per patient averages only over patients in which the subtype is observed.

Table 5: Per-subtype support and pooled out-of-fold F1 with 95% percentile intervals from 2,000 patient-clustered bootstrap resamples. Cells/patient is the mean among patients in which that subtype is observed. Head and Tail denote the seven most and seven least abundant subtypes; all others are Mid.

Subtype	**Lineage**	**Cells**	**Patients**	**Cells/patient**	**Regime**	**DropCascade F1 [95% CI]**	**XGB F1 [95% CI]**
Erythrocyte	Other	65	6	10.8	Tail	0.253 [0.000, 0.395]	0.133 [0.000, 0.545]
Atypical Memory B	B	781	34	23.0	Tail	0.005 [0.000, 0.014]	0.005 [0.000, 0.014]
CD4 Monocyte-4	Myeloid	1,405	35	40.1	Tail	0.027 [0.006, 0.053]	0.014 [0.000, 0.033]
platelet	Other	1,574	38	41.4	Tail	0.078 [0.016, 0.125]	0.063 [0.015, 0.106]
Tcm	T	2,121	36	58.9	Tail	0.000 [0.000, 0.000]	0.004 [0.000, 0.016]
Memory B-2	B	2,130	33	64.5	Tail	0.017 [0.004, 0.033]	0.011 [0.000, 0.024]
Plasma	B	2,173	40	54.3	Tail	0.611 [0.522, 0.684]	0.599 [0.496, 0.685]
pDC	Myeloid	2,615	40	65.4	Mid	0.029 [0.010, 0.061]	0.019 [0.005, 0.040]
CD14 Monocyte-3	Myeloid	2,977	32	93.0	Mid	0.258 [0.170, 0.306]	0.177 [0.103, 0.211]
CD8 + CTL-3	T	3,244	27	120.1	Mid	0.020 [0.004, 0.035]	0.023 [0.008, 0.043]
GCB	B	3,283	31	105.9	Mid	0.039 [0.007, 0.071]	0.047 [0.001, 0.073]
Cycling T	T	3,667	40	91.7	Mid	0.028 [0.015, 0.043]	0.012 [0.003, 0.022]
CD14 Monocyte-2	Myeloid	3,931	40	98.3	Mid	0.075 [0.046, 0.102]	0.078 [0.056, 0.097]
CD8 + CTL-2	T	4,049	33	122.7	Mid	0.069 [0.017, 0.116]	0.078 [0.015, 0.113]
CD16 Monocyte-2	Myeloid	5,142	40	128.6	Mid	0.111 [0.064, 0.161]	0.117 [0.063, 0.164]
CD8 + CTL-1	T	7,303	40	182.6	Mid	0.009 [0.000, 0.024]	0.002 [0.001, 0.004]
cDC2	Myeloid	7,376	40	184.4	Mid	0.132 [0.099, 0.181]	0.125 [0.079, 0.167]
CD8 Tem-2	T	10,548	40	263.7	Mid	0.019 [0.005, 0.038]	0.017 [0.008, 0.030]
CD16 Monocyte-1	Myeloid	14,678	40	366.9	Mid	0.152 [0.111, 0.198]	0.118 [0.090, 0.150]
CD4 + Tcm-2	T	16,861	39	432.3	Mid	0.188 [0.111, 0.236]	0.188 [0.136, 0.220]
Naive T	T	17,960	40	449.0	Head	0.131 [0.067, 0.183]	0.149 [0.073, 0.199]
Memory B-1	B	23,722	40	593.0	Head	0.034 [0.020, 0.048]	0.032 [0.017, 0.047]
CD8 Tem-1	T	26,714	40	667.9	Head	0.187 [0.128, 0.250]	0.162 [0.107, 0.221]
CD14 Monocyte-1	Myeloid	29,871	40	746.8	Head	0.437 [0.378, 0.495]	0.451 [0.392, 0.507]
CD56 NK	NK	38,442	40	961.0	Head	0.398 [0.320, 0.436]	0.437 [0.373, 0.480]
CD16 NK	NK	58,507	40	1462.7	Head	0.383 [0.326, 0.434]	0.376 [0.317, 0.429]
CD4 + Tcm-1	T	70,653	40	1766.3	Head	0.417 [0.368, 0.460]	0.435 [0.393, 0.473]

For continuity with common long-tail reporting, Table 6 aggregates the seven least and seven most abundant subtypes. This summary is secondary to the per-class audit because individual subtypes can depart sharply from the abundance trend.

Model	**Tail Macro-F1**	**Head Macro-F1**
DropCascade	0.142	0.284
Flat FT-style Transformer	0.117	0.290
FT-style Transformer + HCE	0.125	0.288
XGBoost	0.119	0.292
Logistic Regression	0.114	0.197
Linear SVM	0.082	0.204
Random Forest	0.072	0.245
Reference Correlation	0.060	0.114

Table 6: Fine-subtype macro-F1 on the seven least and seven most abundant subtypes under pooled patient-disjoint evaluation. This aggregate is reported as a descriptive complement to the complete per-subtype audit in Table 5.

Cross-lineage error relative to fixed-marginal nulls. Table 7 separates raw error severity from the cross-lineage rate expected after random reassignment of the same predicted subtype labels. For the four main models, observed rates are 0.743–0.757 versus fixed-marginal expectations of 0.671–0.675, yielding negative normalized retention. Because an unconstrained permutation can create chance exact matches, we also report a wrong-conditioned sensitivity null; its 0.741–0.745 range is close to, but not above, the observed rates. Thus the raw 75% cross-lineage finding does not show that these models preserve more hierarchy than their class marginals predict.

Model	Errors	Observed	Null	Wrong-cond. null	Obs.−Null [95% CI]	Norm. retention
DropCascade	251,354	0.750	0.675	0.744	+0.075 [+0.048, +0.098]	-0.111
Flat FT-style Transformer	247,798	0.757	0.675	0.745	+0.082 [+0.055, +0.105]	-0.122
FT-style Transformer + HCE	250,944	0.743	0.671	0.741	+0.072 [+0.045, +0.094]	-0.107
XGBoost	244,990	0.754	0.672	0.741	+0.082 [+0.053, +0.106]	-0.121
Logistic Regression	301,566	0.705	0.720	0.752	-0.015 [-0.031, +0.003]	+0.021
Linear SVM	296,904	0.735	0.738	0.774	-0.003 [-0.022, +0.015]	+0.004
Random Forest	246,420	0.809	0.685	0.749	+0.124 [+0.095, +0.144]	-0.180
Reference Correlation	328,596	0.775	0.758	0.785	+0.018 [+0.001, +0.034]	-0.023

Table 7: Cross-lineage error against fixed-marginal nulls. Null permutes predicted subtype labels within each model's errors while preserving true- and predicted-subtype marginals; its expectation is exact. Wrong-cond. null renormalizes the predicted marginal after excluding each true subtype, preventing chance exact matches but no longer preserving the realized prediction marginal exactly. Confidence intervals for Observed−Null use 2,000 patient-clustered bootstrap resamples. Normalized retention is $(\mathrm{Null} - \mathrm{Observed})/\mathrm{Null}$; positive values indicate fewer cross-lineage errors than expected.

A.5 Protocol Sensitivity: Patient-Level versus Cell-Level Splitting

Table 8 compares the primary patient-disjoint protocol with a leakage-prone cell-level split control. It is placed in the appendix because the central result uses patient-disjoint evaluation throughout; the control quantifies how much the easier protocol would inflate both absolute performance and the apparent advantage of a model family.

Task	Model	Patient-disjoint	Cell split	Inflation
Coarse lineage	XGBoost	0.322	0.372	$+0.050$
	DropCascade	0.326	0.417	$+0.090$
Fine subtype	XGBoost	0.143	0.212	$+0.069$
	DropCascade	0.152	0.260	$+0.108$

Table 8: Macro-F1 under the primary patient-disjoint protocol versus a leakage-prone cell-level split control. Cell-level splitting inflates performance for both methods, and does so more strongly for DropCascade than for XGBoost. This demonstrates that evaluation protocol can change both absolute performance and the apparent relative advantage of model families, which is why APT-Bench treats patient-disjoint evaluation as a requirement rather than a reporting option.

Patient-balanced cell-count sensitivity. Patient-disjoint evaluation prevents leakage but does not make a pooled cell-level point estimate patient-balanced: patient cell counts range from 1,929 to 22,627. Table 9 therefore recomputes both hierarchy endpoints after sampling 1,900 cells without replacement from every patient for each of 50 seeds. Coarse macro-F1 changes by at most $+0.005$, while fine macro-F1 changes by -0.007 to -0.009; the qualitative model ordering is unchanged. The main conclusions are therefore not driven by a few high-cell-count patients, although the headline pooled metric remains explicitly cell-weighted.

Task	Model	Pooled M-F1	Equal-cell M-F1 [95% interval]	Δ
Coarse lineage	DropCascade	0.326	0.331 [0.325, 0.337]	+0.005
	Flat FT-style Transformer	0.323	0.326 [0.322, 0.330]	+0.003
	FT-style Transformer + HCE	0.333	0.334 [0.328, 0.340]	+0.000
	XGBoost	0.322	0.324 [0.320, 0.329]	+0.002
Fine subtype	DropCascade	0.152	0.144 [0.138, 0.153]	-0.008
	Flat FT-style Transformer	0.147	0.138 [0.131, 0.147]	-0.008
	FT-style Transformer + HCE	0.151	0.142 [0.137, 0.151]	-0.009
	XGBoost	0.143	0.136 [0.131, 0.144]	-0.007

Table 9: Patient-balanced cell-count sensitivity. For each of 50 seeds, 1,900 cells are sampled without replacement from every patient, and pooled macro-F1 is recomputed on the resulting 76,000 cells. All models use identical sampled row indices. The interval is the empirical 2.5th–97.5th percentile across seeds; Δ is the equal-cell mean minus the original pooled metric. This diagnostic changes evaluation weights only and does not retrain or reselect models.

A.6 Patient-Level Lineage-Resolved Attribution Audit

We revisit disease attribution at the patient, rather than cell, level. For patient p, lineage m, and aptamer j, we compute $\bar{x}_{pmj} = n_{pm}^{-1} \sum_{i \in (p,m)} x_{ij}$, so every point in Figure 3 represents one patient regardless of cell count. We show the three aptamers with the best mean rank across the five outer-fold PSAS rankings (APT-179, APT-225, and APT-164) in the three major immune compartments. The raw view is paired with a patient-centered view obtained by subtracting each patient's per-aptamer median across all of their cells before lineage aggregation. Disease groups contain only 6–8 patients, and the two views use the same vertical scale.

This is an exploratory full-cohort visualization, not an independently validated biomarker analysis. Raw patient means separate several disease groups, but much of that separation contracts toward zero after patient centering, consistent with the centering sensitivity in Appendix A.3. The available feature manifest contains only anonymized APT identifiers and no verified protein-target mapping. We therefore make no molecular-plausibility, protein-target, or pathway claim from these panels.

A.7 A Euclidean Heuristic Inspired by Hyperbolic Hierarchy Embedding

Motivated by the observation that tree-structured label hierarchies can be embedded with lower distortion under hyperbolic geometry than Euclidean geometry (Nickel & Kiela, 2017), we explored a lightweight Euclidean heuristic that directly enforces hierarchy-consistent distances between subtype representations, without the optimization overhead of a full hyperbolic parameterization (e.g., Riemannian optimization over a Poincaré ball). For each training batch we compute the mean encoder representation (§4.1) of every subtype present in the batch as a prototype, and add a margin-based regularizer that penalizes same-lineage subtype prototypes farther apart than a margin $m_{\text{intra}} = 1.0$ and different-lineage subtype prototypes closer together than a margin $m_{\text{inter}} = 2.0$, added to the training objective (§4.4) with weight 0.1.

Under the single patient-independent split used for architecture development (§3.3), this regularizer did not improve performance and in fact reduced it relative to the same configuration without the term: fine-subtype macro-F1 decreased from 0.171 to 0.136, and coarse-lineage macro-F1 decreased from 0.360 to 0.317. We did not conduct a hyperparameter sweep over the margin values or loss weight, so this diagnostic cannot determine whether the result reflects redundancy, optimization conflict, or an unsuitable configuration. We report the negative single-split result rather than omitting it, but do not treat it as a component ablation or evidence for the necessity of any retained DropCascade component.

A.8 Coarse–Fine Consistency Diagnostic

Independent coarse and fine classifiers can disagree about the hierarchy even when each prediction is plausible in isolation. Table 10 reports their natural agreement. Unlike an earlier single-split

Figure 3: Patient-balanced lineage-resolved distributions for the three highest-ranked stable PSAS aptamers. Top: raw patient-lineage means. Bottom: means after subtracting each patient's per-aptamer median. Each dot is one patient and boxes summarize patients, not cells; group sizes are shown on the horizontal axis. Corresponding raw and centered panels share the same vertical scale. Aptamer selection and visualization use the full cohort and are exploratory.

analysis, this version uses the same five-fold pooled out-of-fold prediction artifacts as the main hierarchy results, so every patient contributes exactly once to both evaluations.

Model	Coarse–Fine Consistency
DropCascade	0.812
Flat FT-style Transformer	0.795
FT-style Transformer + HCE	0.834
XGBoost	0.736
Logistic Regression	0.568
Linear SVM	0.631
Random Forest	0.573
Reference Correlation	0.380

Table 10: Natural coarse–fine consistency on the same 361,792-cell, 40-patient pooled out-of-fold artifacts as Table 2. For each model, the hard fine prediction is mapped to its parent and compared with the corresponding hard coarse prediction. *Unknown* is excluded. Consistency is supporting evidence rather than predictive superiority because it can be changed by constrained decoding.

A.9 Minimal Aptamer Panel for Disease Classification

Because a smaller aptamer panel would be cheaper and simpler to deploy clinically, we asked how many of the 293 aptamers are needed to preserve the observed within-cohort disease result. Patient-Stable Aptamer Selection (PSAS) denotes the fold-specific ranking and development-only budget-selection protocol below; "stable" refers to comparison of ranks across outer folds, not to patient-balanced fitting of the LR ranker.

PSAS protocol.

1. **Outer isolation.** For outer fold o, the eight test patients are sealed; the other 32 patients form the outer-development set. Feature standardization, ranking, budget selection, and model fitting use no outer-test cells.
2. **LR ranking score.** On all cells in the outer-development set, each aptamer is standardized using development-only mean and variance. We fit an L1-regularized six-class multinomial LR (SAGA, $C = 0.1$) and rank aptamer j by $s_j = \sum_{d=1}^{6} |\beta_{dj}|$, the sum of its absolute coefficients over disease classes. Cells are not first aggregated by patient and receive equal weight, so this ranking is cell-count weighted.
3. **Development folds.** The 32 development patients are shuffled with the seeded pipeline RNG and divided into five patient-disjoint, non-stratified inner folds. The outer-development ranking is held fixed; for each $B \in \{5, 10, 20, 40, 80, 160, 293\}$, an LR is refitted on each inner-training split using its Top-B features and scored by patient-majority-vote macro-F1 on the corresponding inner validation patients.
4. **Budget rule.** Let $\bar{F}_B$ be mean inner-fold patient macro-F1. We select $B^* = \min\{B : \bar{F}_B \geq \bar{F}_{293} - 0.02\}$. The absolute margin $\delta = 0.02$ was fixed in the analysis code before outer-test scoring, but was not prospectively preregistered. Because ranking is fixed from all outer-development patients rather than refitted inside each inner split, this is development-only budget selection conditional on one ranking, not fully nested inner re-ranking.
5. **Outer evaluation.** The fold-specific Top-B^* features are taken from that outer-development ranking. LR and XGBoost are then trained from scratch on all outer-development cells restricted to those features, with standardization refitted there, and evaluated once on the outer-test patients. B^* is selected using LR only; XGBoost in Table 11 and Figure 1 evaluates the LR-selected features.
6. **Random controls.** For each $B \in \{5, 10, 20\}$, 100 feature sets are sampled uniformly without replacement. One sampled set per repeat is shared across the five outer folds, while LR and XGBoost are independently refitted within every fold. Thus panels are independent across repeats, not independently redrawn across outer folds.
7. **SHAP control.** The separate XGBoost-SHAP analysis fits a full-panel XGBoost model in each outer-development set, forms a patient-balanced mean absolute TreeSHAP ranking,

and evaluates both LR and XGBoost on either the LR-selected or SHAP-selected Top-10/20 features. SHAP features do not determine B^* or the main curve.

All five outer folds select $B^* = 10$ under this rule.

Budget terminology. An *aptamer measurement budget* B means that a model is retrained and evaluated using only those B input features. An *inference-time cell budget* C means that, after training on all available outer-development cells, only C cached predictions are sampled from each held-out patient before majority-vote aggregation. We do not reduce or evaluate a *training-time cell budget*; the cell-count sensitivity therefore supports low-cell inference aggregation, not training with only C cells per patient.

The resulting development-selected 10-aptamer panels reproduce the full-panel patient-level macro-F1 of 1.0 for both Logistic Regression and XGBoost on all five outer test folds. Figure 1 compares these selected panels with 100 Random-B repeats at each compact budget. For every model and budget, we report the random-panel mean, empirical 2.5th–97.5th percentile interval, and the empirical midrank percentile of Top-B within that distribution (Table 12).

Outer Fold	δ	B^*	LR Macro-F1	XGBoost Macro-F1
0	0.02	10	1.000	1.000
1	0.02	10	1.000	1.000
2	0.02	10	1.000	1.000
3	0.02	10	1.000	1.000
4	0.02	10	1.000	1.000
All folds	**0.02**	**10**	**1.000**	**1.000**

Table 11: Outer-test-isolated patient-level panel selection for disease recognition. In each outer fold, aptamer ranking and budget selection are performed using outer-development patients only, and the selected budget is evaluated once on the untouched outer-test patients. The ranking is fixed across inner validation splits; B is selected conditional on that ranking. Under the analysis-specified, non-preregistered tolerance margin $\delta = 0.02$, all five folds select the same development budget $B^* = 10$, and the resulting 10-aptamer panels preserve the observed full-panel patient macro-F1 for both Logistic Regression and XGBoost. This result supports a measurement-efficiency interpretation rather than a claim of unique causal biomarkers.

B	Model	Top-B	Random mean [95% interval]	Percentile
5	LR	0.898	0.663 [0.422, 0.890]	97.0
5	XGBoost	0.925	0.725 [0.453, 0.890]	99.0
10	LR	1.000	0.868 [0.670, 0.952]	100.0
10	XGBoost	1.000	0.921 [0.781, 1.000]	97.0
20	LR	1.000	0.973 [0.900, 1.000]	79.5
20	XGBoost	1.000	0.981 [0.923, 1.000]	75.0

Table 12: Compact-panel random controls over 100 panel repeats. Top-B uses fold-specific outer-development LR rankings. Each repeat draws one random panel shared across outer folds; both classifiers are refitted in every fold. The percentile is the empirical midrank of Top-B within the random distribution.

At $B = 10$, the Top-B panel is at the 100th empirical percentile for LR and the 97th for XGBoost; Random-B means are 0.868 and 0.921, respectively. By $B = 20$, random means rise to 0.973 and 0.981 and many random panels also saturate, reducing the corresponding midrank percentiles to 79.5 and 75.0. Thus ranking is most consequential at the smallest budgets, while the broader panel contains extensive substitutable signal.

We next crossed aptamer budget $B \in \{5, 10, 20, 293\}$ with inference-time cell caps $C \in \{100, 500, 1000, 5000, \mathrm{all}\}$. For each finite cap, cached predictions were sampled without replacement within each outer-test patient; patient majority votes were pooled across the five held-out folds, and the process was repeated over 50 sampling seeds. Training still used all outer-development cells.

At $B = 5$, macro-F1 varies with the sampled cells and reaches 0.881 on average at $C = 100$; at $B = 10$, 20, and 293, macro-F1 is 1.0 for every cap and every seed (Table 13).

B	100 cells	500 cells	1,000 cells	5,000 cells	All cells
5	0.881 [0.845, 0.925]	0.894 [0.852, 0.925]	0.906 [0.875, 0.925]	0.899 [0.898, 0.898]	0.898 [0.898, 0.898]
10	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]
20	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]
293	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]

Table 13: LR patient-level macro-F1 under inference-time equal-cell subsampling. Entries are mean [empirical 95% interval] over 50 seeds; cached outer-test predictions are subsampled before aggregation. Models are trained once using all outer-development cells, so this is not a training-time cell-budget experiment.

Finally, we tested whether the compact result depends on Logistic Regression coefficients as the attribution mechanism. Within each outer-development set, we fitted a full-panel XGBoost model and computed absolute TreeSHAP contributions on at most 1,000 cells per development patient. We first averaged over cells within each patient and then over patients, preventing high-cell-count patients from dominating the attribution ranking. The LR–SHAP Jaccard overlap is moderate rather than complete (0.501 for Top-10 and 0.475 for Top-20), yet panels from either ranking achieve patient macro-F1 1.0 when evaluated with either LR or XGBoost (Table 14). This cross-model control supports measurement efficiency and predictive redundancy, not a unique molecular signature.

B	LR–SHAP Jaccard	Ranking	LR eval.	XGB eval.
10	0.501 ± 0.126	LR coefficient	1.000	1.000
10	–	XGBoost-SHAP	1.000	1.000
20	0.475 ± 0.087	LR coefficient	1.000	1.000
20	–	XGBoost-SHAP	1.000	1.000

Table 14: Cross-model attribution control. Jaccard is mean $\pm$ SD across five outer-fold Top-B sets. LR-coefficient and XGBoost-SHAP rows use their own selected features; evaluation columns report pooled patient macro-F1 after refitting LR or XGBoost. SHAP ranking does not select B^*.

The fold-overlap LR ranking is also stable. The highest-ranked aptamers by mean outer-fold rank are APT-179, APT-225, APT-164, APT-166, APT-57, APT-69, APT-125, APT-161, APT-172, and APT-218, and 290 of the 293 aptamers are retained after a simple redundancy screen at $|r| \leq 0.9$. Appendix A.6 visualizes the top three at the patient-lineage level, including a patient-centering control; because only anonymized IDs are available, neither analysis supports protein-target or pathway interpretation.

A.10 Balanced Sampling Ablation

The final DropCascade configuration (§4.4) does not use class-balanced sampling. We initially implemented a balanced subtype-disease batch sampler to counteract the long-tailed subtype distribution (§3.4), but found and fixed an implementation issue in which `steps_per_epoch` was computed from the natural data loader's batch size (512) rather than the sampler's own, much smaller, per-batch cell count (e.g., 14 subtypes $\times$ 2 diseases $\times$ 4 cells $= 112$); this mismatch caused each nominal epoch to cover only about 22% of the training set when the sampler was enabled, silently under-training the model and desynchronizing every epoch-indexed schedule (warmup, contrastive ramp-up, routing ramp-up, early-stopping patience) relative to the sampler-disabled baseline.

After fixing this issue, the balanced sampler improved fine-subtype macro-F1 from 0.102 to 0.119 on the single patient-independent split used for architecture development, but both remain below the 0.171 achieved without any balanced sampling at all. We report this ablation as a diagnostic finding about our specific sampler implementation, not as evidence against balanced sampling in general, and did not further redesign the sampler given this result; the final configuration therefore omits it.

Held-out Disease	Fine Macro-F1	Fine Acc.	Coarse Macro-F1	Coarse Acc.	Tail Macro-F1
BTC	0.086	0.317	0.310	0.479	0.073
CRC	0.105	0.277	0.309	0.520	0.112
GC	0.063	0.269	0.281	0.488	0.057
HCC	0.063	0.246	0.264	0.383	0.072
PC	0.057	0.253	0.221	0.482	0.037
Macro-average	**0.075**	**0.272**	**0.277**	**0.470**	**0.070**

Table 15: Leave-one-disease-out generalization for DropCascade over the five malignant held-out diseases. Each row withholds all patients of the named disease from training. The in-distribution reference using the same model under the standard patient-disjoint 5-fold protocol is fine Macro-F1 0.152 and coarse Macro-F1 0.326. A separate Normal-held-out run produces near-chance fine performance. LODO is reported only as an exploratory appendix stress test, not as proof that the model already solves cross-disease generalization.

Held-out Disease	Proxy-Sup. Head AUROC	Proxy-Sup. Head AUPR	Energy AUROC
BTC	0.782	0.855	0.591
CRC	0.874	0.901	0.779
GC	0.729	0.698	0.575
HCC	0.881	0.842	0.558
Normal	0.607	0.843	0.722
PC	0.999	0.999	0.539
Macro-average	**0.812**	**0.856**	**0.627**

Table 16: Exploratory novelty detection under the nested leave-one-disease-out protocol. For each held-out disease, the proxy-supervised head is learned on a proxy-novel disease drawn from the remaining training diseases and then evaluated on the truly held-out disease, which is unseen during both cascade and novelty-head training. Energy is a training-free confidence score and does not receive matched proxy-novel supervision, so this table is descriptive rather than a fair method comparison. AUPR additionally depends on the held-out-disease prevalence in each evaluation set.

A.11 Exploratory Disease-Level Novelty Track

We treat disease-level novelty as an exploratory appendix track rather than a main contribution. In this protocol, all patients from one disease category are withheld during training and evaluated as a disease-level distribution shift. This is useful for stress-testing representations, but it is not a fair basis for method ranking because the cohort contains only six disease groups, proxy-disease supervision is limited, and the supervised detector and training-free confidence scores do not receive matched novelty supervision.

Table 16 reports two standard training-free scores, maximum softmax probability (Hendrycks & Gimpel, 2016) and energy (Liu et al., 2020), alongside a proxy-supervised novelty head. For each held-out disease, the cascade is trained only on the known diseases. The encoder is then frozen, and a lightweight MLP novelty head is trained to distinguish one proxy-novel known disease from the remaining known diseases. The truly held-out disease is never used during either cascade training or novelty-head training.

Under this exploratory nested leave-one-disease-out protocol, the proxy-supervised head attains higher descriptive AUROC than the training-free confidence scores. This should be interpreted narrowly: proxy disease supervision can help separate a withheld disease from known diseases in this cohort. It does not establish that the novelty head is intrinsically better than energy scoring, because the comparison lacks matched supervised probes on raw APT features, classical classifiers, patient-clustered uncertainty intervals, proxy-choice sensitivity, and a formal ID/OOD patient manifest.